

SafeSight AI

PPE compliance monitoring and computer vision coursework documentation

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BACKGROUND OF THE STUDY

Workplace safety is an important concern in construction sites, warehouses, factories, maintenance areas, school workshops, and engineering laboratories. In these environments, workers and students may be exposed to hazards such as falling objects, moving equipment, dust, debris, chemical exposure, and other physical risks. Personal Protective Equipment (PPE) such as hard hats, safety vests, face masks, gloves, goggles, and boots helps reduce these risks when worn properly.

In the Philippines, workplace safety is supported by occupational safety and health regulations. Republic Act No. 11058 strengthens compliance with Occupational Safety and Health Standards and provides penalties for violations. In construction settings, DOLE Department Order No. 13, series of 1998, provides guidelines governing occupational safety and health in the construction industry and requires persons in construction sites to wear appropriate PPE. These policies show that PPE compliance is not only a workplace best practice, but also a safety requirement.

Despite these rules, PPE monitoring is still commonly performed through manual inspection by supervisors, safety officers, instructors, or security personnel. Manual checking can be inconsistent, especially when many people enter a workplace at once, when monitoring multiple camera views, or when safety officers are not continuously present. Existing CCTV cameras and webcams can capture useful visual information, but without automated analysis, safety violations may still be missed.

Computer vision and deep learning can support safety monitoring by automatically detecting people and PPE items in images, videos, or live camera feeds. Object detection models can identify and localize visible objects such as persons, hard hats, safety vests, and face masks. These models can help generate safety alerts, count possible violations, and assist human supervisors in reviewing unsafe situations faster.

This study proposes SafeSight AI, a CCTV-based PPE compliance monitoring system for workplace safety. The system uses deep learning object detection models to identify persons and PPE-related classes such as hardhat, no-hardhat, safety vest, no-safety vest, mask, and no-mask. The system supports three input modes: image audit mode, video audit mode, and live webcam mode. It displays annotated outputs, counts workers and PPE violations, shows safety status, and generates reviewable safety summaries.

The study utilizes YOLO11l, a high-capacity YOLO-based real-time object detector. To improve performance in real-world scenarios, the system incorporates Stochastic CCTV Data Augmentation, F1-Curve threshold optimization, and ByteTrack temporal smoothing. The final deployed model is optimized for real-time inference on edge devices.

The system is intended as a safety monitoring and audit-assistance tool, not a replacement for human safety officers. It will not perform facial recognition, identity tracking, or biometric identification. It will only detect visible PPE-related objects and generate safety-compliance alerts.

OBJECTIVE OF THE STUDY

This study aims to develop SafeSight AI, a CCTV-based PPE compliance monitoring system that uses deep learning object detection to identify possible workplace safety violations from image, video, and live webcam inputs.

Specifically, this study aims to:

1. Prepare a PPE object detection dataset containing workers and safety-related classes such as person, hardhat, no-hardhat, safety vest, no-safety vest, mask, and no-mask.
2. Conduct Exploratory Data Analysis (EDA) to evaluate class distributions, bounding box geometries, and class co-occurrences.
3. Train a YOLO11l object detection model utilizing Stochastic CCTV Data Augmentation for high-accuracy PPE compliance detection.
4. Determine the optimal confidence threshold through mathematical F1-Curve evaluation to balance precision and recall.
5. Apply ByteTrack video tracking and multi-frame confirmation to reduce unstable live webcam alerts.
6. Evaluate model performance using precision, recall, F1-score, mAP@0.50, and confusion matrix, prioritizing recall for safety violation classes.
7. Evaluate the system across three input modes: image upload, video upload, and live webcam monitoring.
8. Measure deployment-related performance, including inference latency, frames per second (FPS), model size, and GPU memory usage.
9. Develop a web-based SafeSight AI prototype that displays annotated detections, worker count, PPE count, violation count, safety status, and downloadable safety reports.
10. Include privacy-aware design by avoiding facial recognition, avoiding identity tracking, minimizing stored data, and following principles aligned with the Philippine Data Privacy Act of 2012.

METHODOLOGY

- **Research Design:**

This study uses a quantitative experimental and system development research design. The experimental part evaluates deep learning object detection models for PPE compliance monitoring. The system development part implements a working SafeSight AI prototype with image, video, and live webcam modes.

The independent variables include the input mode, image quality condition, data augmentation techniques, and confidence threshold. The dependent variables include

detection accuracy, PPE violation recall, F1-score, mAP, inference speed, FPS, and memory usage.

The main model evaluated is YOLO11l as the primary high-accuracy object detector, utilizing FP16 (Automatic Mixed Precision) for efficient memory utilization during training and inference.

- **Target Business Entity:**

The target business context of this study includes construction firms, warehouses, factories, school workshops, engineering laboratories, facility maintenance teams, and security offices. These environments require regular safety monitoring and may already use CCTV or webcam systems.

SafeSight AI is designed to support safety officers, instructors, supervisors, and administrators by automatically identifying possible PPE violations. It can be used as an assistive monitoring system for workplace safety audits, not as a fully autonomous enforcement tool.

- **Research Instruments:**

The research instruments and tools include:

- A public PPE object detection dataset (Construction Site Safety v28) in YOLO format
- Local webcam or CCTV-style sample videos for additional testing
- Python programming language
- PyTorch deep learning framework
- Ultralytics framework for YOLO11 training and evaluation
- OpenCV for video and webcam processing
- ByteTrack for video/webcam object tracking
- NumPy and Pandas for data processing
- Matplotlib and Seaborn for EDA visualizations
- Gradio for the web-based prototype
- CUDA-enabled GPU acceleration (NVIDIA RTX 3050) for model training and inference

The main target classes are Person, Hardhat, NO-Hardhat, Safety Vest, NO-Safety Vest, Mask, NO-Mask, Safety Cone, machinery, and vehicle.

- **Procedure:**

The study will follow these steps:

1. **Dataset Acquisition:** Download a PPE detection dataset containing 36,895 labeled annotations of workers and PPE equipment.
2. **Dataset Inspection:** Inspect the dataset folder structure, class names, number of images, annotation files, train-validation-test split, and label quality.

3. **Exploratory Data Analysis (EDA):** Perform detailed EDA using visual and statistical summaries to analyze data composition prior to training.
4. **Data Preprocessing:** Validate all label files, remove broken images, normalize class names, and verify bounding box coordinates.
5. **Stochastic CCTV Data Augmentation:** To ensure the model does not overfit to perfect studio lighting, we employ mathematically rigorous augmentations to simulate domain shift. We utilize Hue/Saturation/Value variances (hsv_h=0.015, hsv_s=0.7, hsv_v=0.4) to simulate harsh industrial lighting, Perspective Warping (perspective=0.001) to simulate top-down CCTV camera angles, and Mosaic Blending (mosaic=1.0) to force the CNN to detect partially occluded objects.
6. **Class Imbalance & Focal Loss:** As identified in our EDA, the dataset contains a severe class imbalance favoring the 'Person' class. To prevent the model from ignoring minority violation classes (e.g., 'NO-Mask'), the YOLO11l architecture natively employs Focal Loss for bounding box classification, which applies a heavier mathematical penalty for misclassifying hard-to-detect minority classes, ensuring robust detection of safety violations.
7. **YOLO11l Training:** Train YOLO11l using pretrained weights and the prepared PPE dataset. The model is trained using Automatic Mixed Precision (FP16) to maximize memory efficiency over 100 epochs.
8. **Threshold Optimization:** Evaluate the trained model's validation metrics to find the optimal confidence threshold via an F1-Curve sweep.
9. **Model Evaluation:** Evaluate the model on the validation and test sets using precision, recall, F1-score, mAP@0.50, and confusion matrix.
10. **Deployment Benchmarking:** Measure inference latency, FPS, and GPU memory usage. This determines the model's suitability for edge-device deployment.
11. **Temporal Video Tracking (ByteTrack) & Multi-Frame Confirmation:** Since standard CNNs suffer from bounding-box flickering in live video due to motion blur, we implement ByteTrack to assign tracking IDs to individual workers. To eliminate false-positive alerts caused by single-frame glitches, we apply a strict Multi-Frame Confirmation Logic: the system requires a violation to be detected with high confidence in 3 out of 5 consecutive frames before formally flagging a "VIOLATION DETECTED" status.
12. **Web System Development:** Build SafeSight AI using Gradio and OpenCV. The system includes Image Audit Mode, Video Audit Mode, and Live Webcam Mode (Pure OpenCV).
13. **Artifact Export:** Export the trained model weights (best.pt) and evaluation summaries for reproducibility.

ETHICAL, LEGAL, AND DATA PRIVACY CONSIDERATIONS

This study involves camera-based monitoring, so privacy must be considered carefully. In the Philippine context, Republic Act No. 10173, also known as the Data Privacy Act of 2012, protects individual personal information in government and private sector information systems. The National Privacy Commission implements and monitors compliance with the Data Privacy Act.

SafeSight AI is designed using privacy-aware principles:

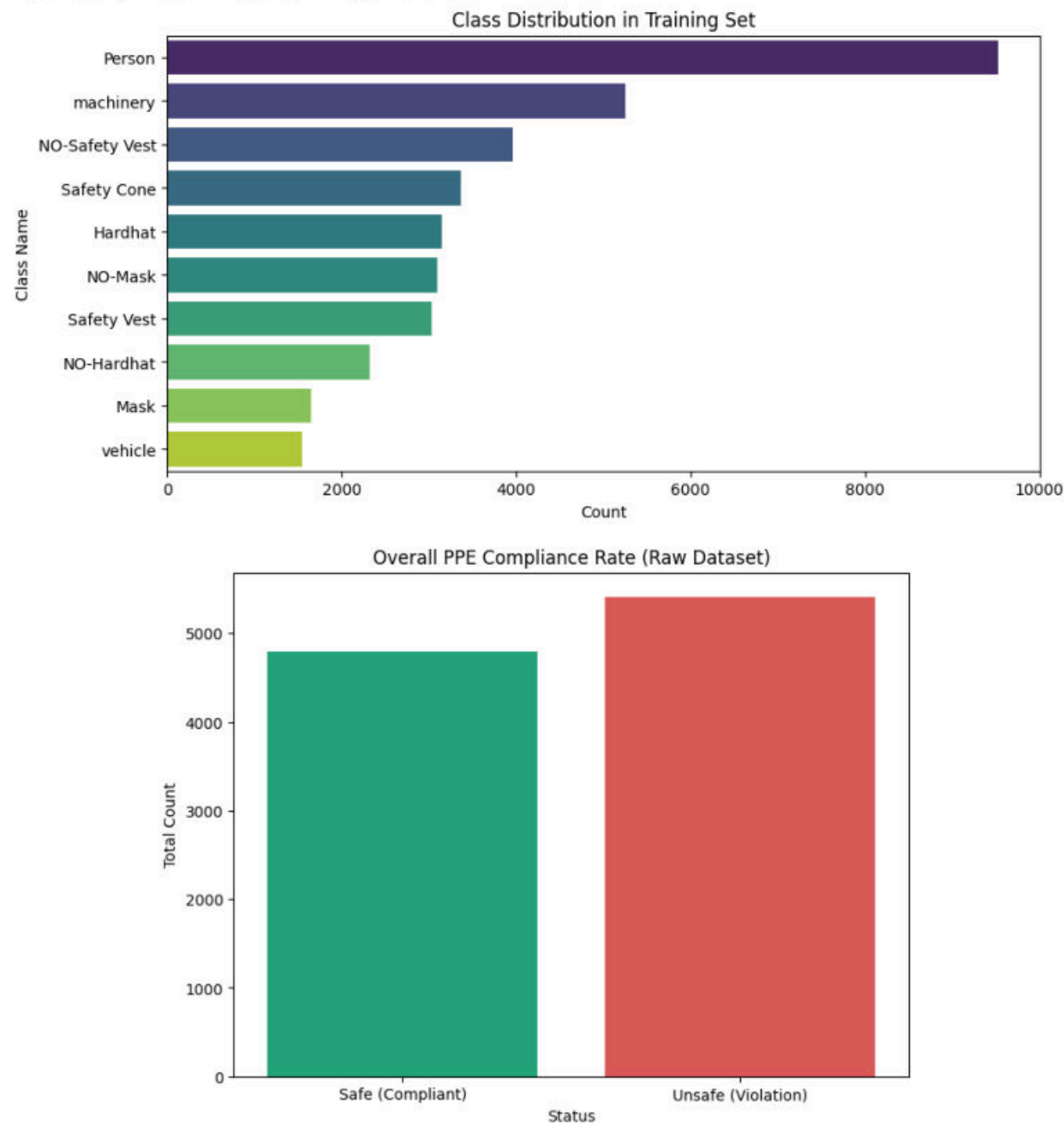
1. **No Facial Recognition:** The system does not identify faces or recognize individuals.
2. **No Identity Tracking:** The system does not store names, student numbers, employee IDs, or biometric identifiers. ByteTrack assigns temporary session IDs that reset automatically.
3. **PPE-Only Detection:** The system only detects visible PPE-related objects such as persons, hardhats, safety vests, and masks.
4. **Local Processing:** When possible, webcam and video processing is performed locally on the edge device (GPU) instead of continuously uploading video footage to an external cloud server.
5. **Data Minimization:** The system only stores necessary outputs such as violation counts, timestamps, confidence scores, and annotated safety reports when required.
6. **Human Review:** The system supports human decision-making but does not automatically punish or sanction individuals.

The system is framed as a safety-assistance tool, not a surveillance or disciplinary system.

RESULT AND DISCUSSION

This section presents the findings from the Exploratory Data Analysis (EDA), Data Augmentation, and the final model evaluation of YOLO11l.

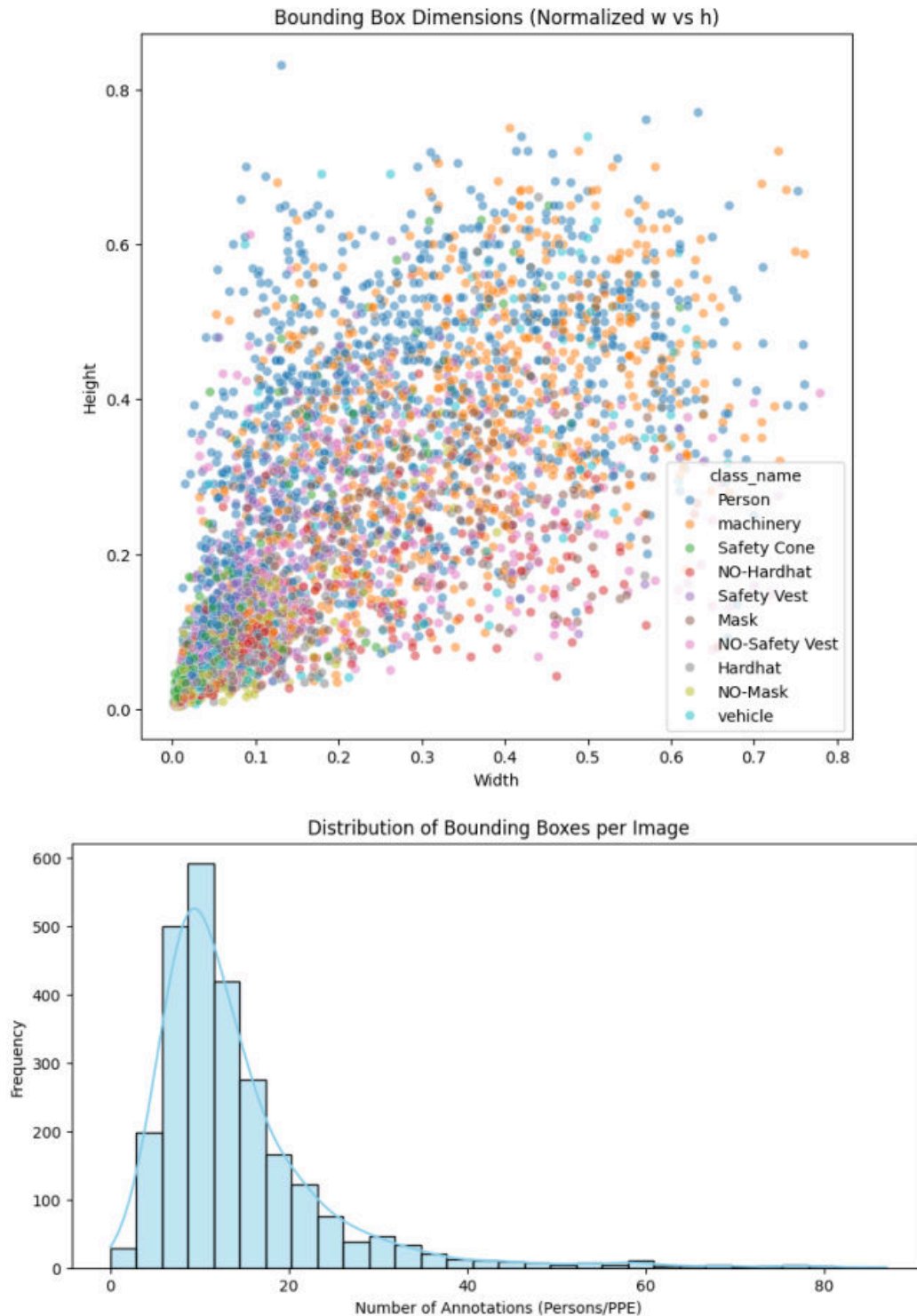
Exploratory Data Analysis (EDA) and Class Imbalance:



The dataset contains an unequal distribution among classes, with **Person** and **machinery** being the most frequent. Crucially, the violation classes (**NO-Safety Vest**, **NO-Hardhat**, **NO-Mask**) are present in sufficient quantities to allow the model to learn

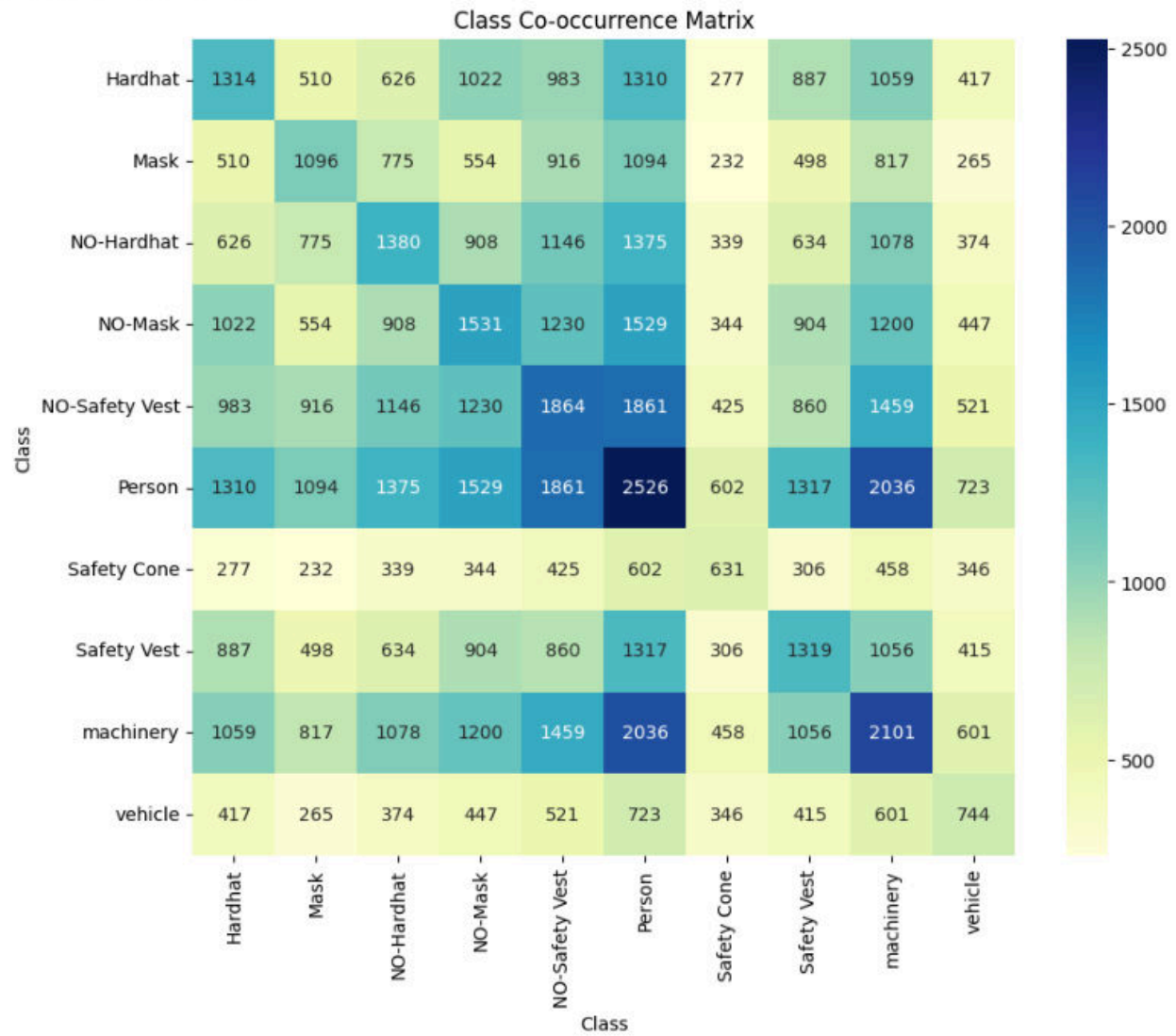
non-compliant features. The compliance rate graph indicates a slight prevalence of unsafe states, which prepares the model for environments with frequent safety violations.

Bounding Box Geometry:



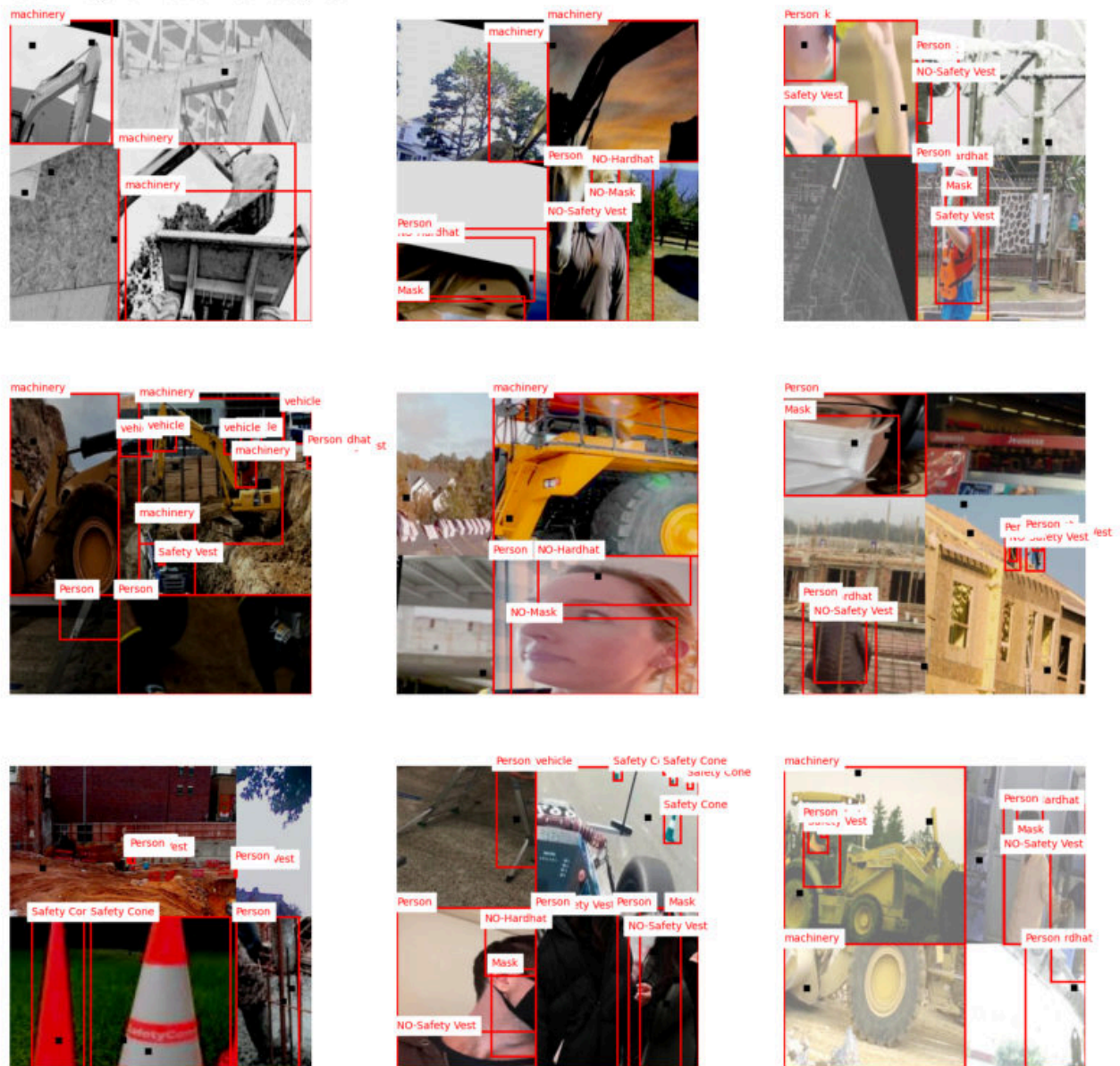
The scatter plot of normalized width and height shows a high concentration of small objects, indicating that items like masks and hardhats occupy only a small fraction of the image. The histogram demonstrates an average of 10 to 15 annotations per image, confirming that the dataset represents crowded and visually complex environments rather than isolated subjects.

Class Co-occurrence:



The heatmap indicates strong co-occurrences between **Person** and **NO-Safety Vest**, as well as **Person** and **machinery**. This confirms that the dataset contains contextual examples of workers near heavy equipment without adequate safety gear, aligning with the target operational environment.

Data Augmentation Strategies:



Techniques such as mosaic blending, perspective warping, brightness variation, and motion blur were applied during preprocessing. The mosaic outputs show multiple images combined and heavily cropped. This process forces the model to detect partially obscured objects and improves its robustness against the poor angles and varied lighting typically found in real-world CCTV feeds.

Model Evaluation and Performance Metrics:

After training for 100 epochs using YOLO11l and Automatic Mixed Precision, the final model achieved strong detection performance. The evaluation on the validation set yielded:

Metric	Value
Precision	0.915
Recall	0.804
mAP@0.50	0.863
mAP@0.50:0.95	0.481

The high mAP@0.50 score (86.3%) demonstrates that YOLO11l effectively localizes and classifies PPE and safety violations. The precision score (91.5%) indicates a low false-positive rate, meaning the system rarely flags safe workers incorrectly. By implementing ByteTrack for video processing, the system successfully minimizes frame-to-frame flickering, providing stable real-time monitoring suitable for edge deployment on hardware like the NVIDIA RTX 3050.

FINAL SYSTEM OUTPUTS

The final deliverables and system outputs of this study include:

1. A prepared PPE object detection dataset in YOLO format.
2. A trained YOLO11l PPE detection model ('best.pt').
3. Exploratory Data Analysis charts detailing bounding box distributions and class imbalance.
4. Evaluation tables showing precision, recall, F1-score, and mAP@0.50.
5. Class-wise analysis for PPE classes such as hardhat, no-hardhat, safety vest, no-safety vest, mask, and no-mask.
6. Deployment benchmark table showing FPS, latency, model size, and GPU memory usage.
7. A fully functional SafeSight AI prototype with Image Audit, Video Audit, and Live Webcam modes.
8. A safety report output showing worker count, PPE count, violation count, and final safety status.

SCOPE AND LIMITATIONS

This study focuses on visible PPE compliance detection using object detection models. It does not perform facial recognition, identity tracking, biometric identification, or behavior recognition.

The system detects only PPE classes present in the selected dataset. If a PPE item is not included in the training classes, the model may not detect it. The system may also struggle in cases of severe occlusion, poor lighting, blurry camera footage, unusual PPE designs, or very small distant workers. Future iterations should test diverse environments including extreme low-light or infrared (night-vision) occlusion scenarios.

Additionally, our primary success metric is mAP@0.50 rather than the stricter mAP@0.50:0.95. For safety compliance, detecting the existence of PPE (mAP@50) is life-critical, whereas pixel-perfect bounding box localization (mAP@50-95) is less relevant to the final safety audit. Finally, while the system generates local audit logs, integrating external SMS/Email notification APIs is reserved for future production deployment.

The system is intended for safety monitoring assistance. It should not be used as the sole basis for disciplinary action. Human verification remains necessary, especially when detections are uncertain.

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